EDA - Exploratory Data Analysis:

- It's the practice of exploring your dataset, by utilizing visualization tools and statistical measures to understand and extract underlying patterns and information within the data.
- This gives a clearer picture of the data, helps us make data-informed decisions and solve crucial business problems with much fewer assumptions and more facts.
- This step is the backbone of any Data Science project and takes up a major chunk of the project timeline.

Additional Note:

- The term EDA was coined by the late mathematician, John Tukey
- First introduced in his book "*Exploratory Data Analysis*" (1977)
- Mr. Tukey also introduced the "Box Plot"

What is done in EDA stage?

Look at and analyze numbers and plots

- **Descriptive Statistics**
  - Central Tendency
  - Dispersion / Spread
  - Distribution of Variables
  - Symmetry and Shape of variables

- **Inferential Statistics**:
  - Strength of Association (correlation)
  - Hypothesis Testing

- **Plots / Graphs**:
  - Univariate
  - Bivariate
  - Multivariate

1. **Measure of Location:**
    - Mean
    - Mode
    - Median
    - Percentiles
    - Quartiles

2. **Measure of Spread:**
    - Variance
    - Standard Deviation
    - Coefficient of Variance
    - Mean Absolute Error (MAE)
    - Inter Quartile Range (IQR)
    - Median Absolute Deviation from Median (MAD)

3. **Measure of Symmetry:**
    - Skewness

4. **Measure of Shape:**
    - Kurtosis

1. **Strength of Association:**
    - Pearson's Correlation
    - Spearman's Rank Correlation
    - Cramer's V

2. **Hypothesis Testing:**
    - ***Test for Normality:***
        - Shapiro-Wilk Test
        - Anderson-Darling Test
    - ***Test for Association:***
        - Numeric Variables:
            - Pearson's Test
            - Spearman's Test
        - Categorical Variables:
            - Chi-Square Test
        - Numeric - Categorical Variable:
            - One-way ANOVA Test
            - Kruskal-Wallis Test
    - ***Steps Involved:***

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  - State the hypotheses:

    - *Null Hypothesis*
    - *Alternate Hypothesis*

  - Determine significance level (alpha ~ 5%)

  - Determine which test to perform

  - Collect necessary data (sample)

  - Obtain Critical values

  - Compute Test Statistic (and p-value)

  - Compare:

    - *Significance level vs p-value; or*
    - *Critical values vs Test Statistic*

  - State conclusions

1. Univariate:

- ***Numeric:***
    - Histrogram
    - KDE Plot
    - Rug Plot
    - Box Plot
    - Violin Plot (box plot + kde plot)
    - Q-Q Plot
- ***Categorical:***
    - Count Plot
    - Pie Chart
- ***Time-related:***
    - Line Plot
    - Aggregated Line Plot

2. Bivariate:

- ***Numeric - Numeric:***
    - Scatter Plot
    - Hexagonal Bin Plot
    - Contour Density Plot
- ***Numeric - Categorical:***
    - Bar Plot

- Box Plot
- KDE Plot
- Violin Plot

- ***Categorical - Categorical:***

  - Bar Plot
  - Stacked Bar Plot
  - Frequency Heatmap

3. Multivariate:

- ***Pair Plots***

- ***Correlation Heatmap:***

  - Pearson
  - Spearman's Rank
  - Cramer's V

- ***Facet Grid (Seaborn)***

Sequence of Steps:

- Import libraries
- Read the '*training*' data subset
    - Check data types
    - Fix data types (if applicable)
- Gather high-level summary of the data
    - .info() method
    - .describe() method on numeric and categorical features
- High-level analysis of missing values:
    - Bar plot
    - Count plot
    - Missingno
- High-level analysis of outliers:
    - Isolation Forest
- Pair plots
- Correlation Analysis (heatmaps)
    - Numeric (Pearson's / Spearman's)
    - Categorical (Cramer's V)
- Detailed Analysis of each Feature:
    - Summary

- Univariate plots
- Bivariate plots (w.r.t. the target variable)
- Hypothesis Testing (normality, strength of association)
- Multivariate plots
- Inspect missing values and extreme values in-depth
  - Filter for necessary subsets
  - Inspect values of other features (plots, summary stats)
- Note observations

- Feature Engineering

  - Create new features
  - Repeat above steps for newly created features

- Repeat above steps iteratively

- Note all observations

- **Pandas Profiling (ydata-profiling)**
- **Sweetviz**
- **Autoviz**
- **D-Tale**